SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA510l

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

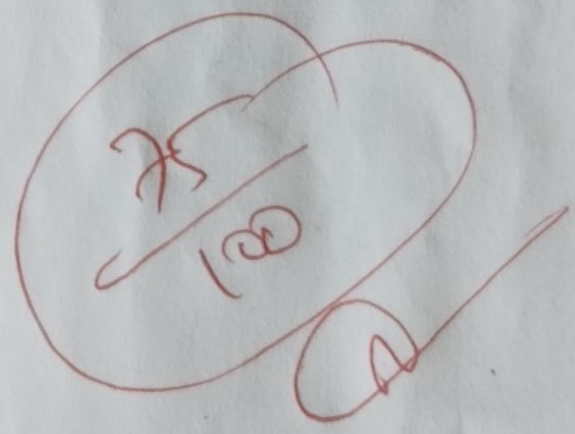
CONCEPT MAPPING

Code of Conduct:

I D. Naveen *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Naveen



Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

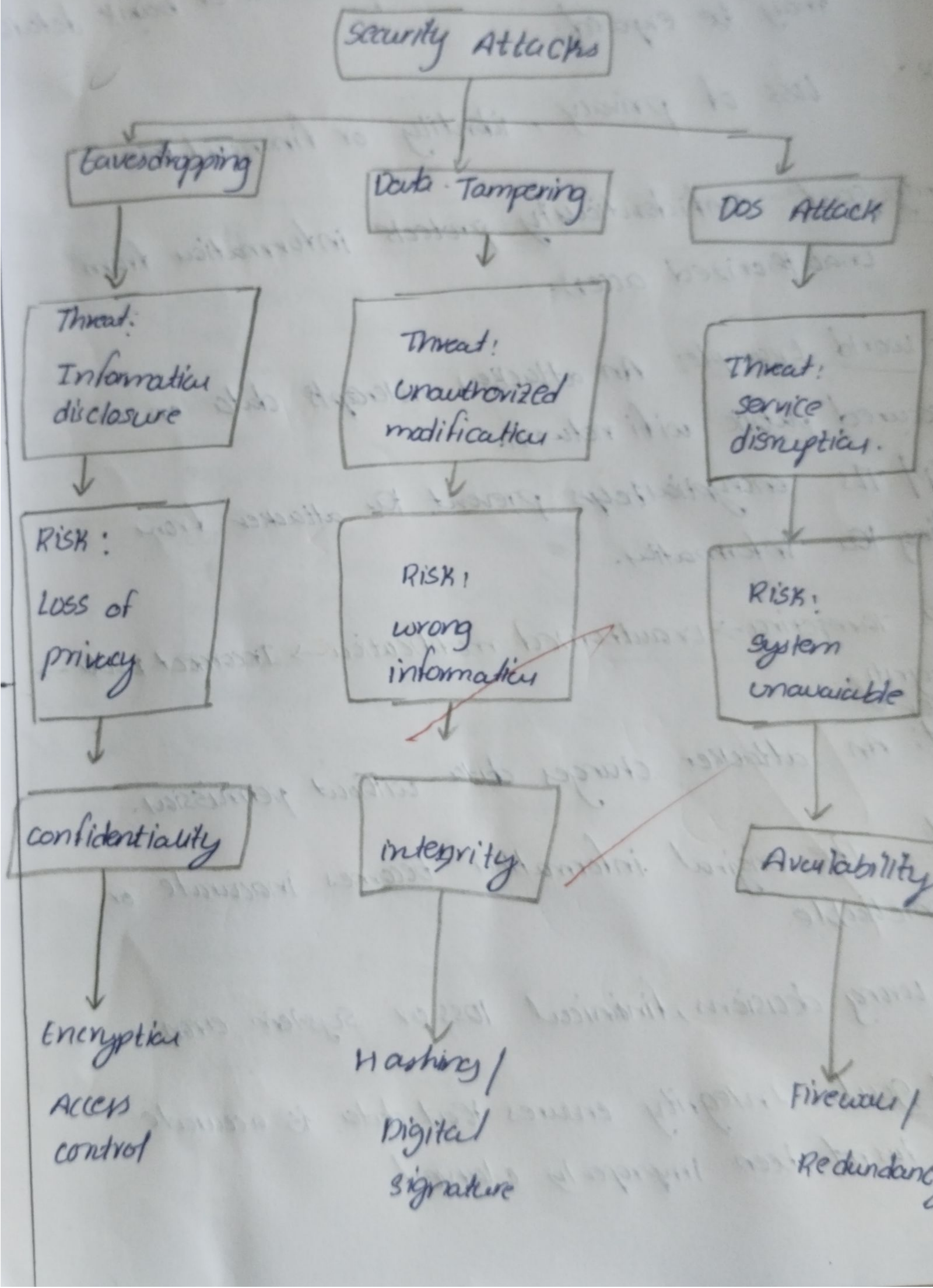
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

75/100

Concept Mapping

CO- Assessment Tool-2

Concept map: Security Attacks, Threats, Risks and security Goals.



1. Eavesdropping → information disclosure → privacy RISK → confiden

Attack! - An attacker secretly listens to network comm

Threat! - sensitive information such as password or bank deta may be exposed.

RISK :- Loss of privacy, identity or financial loss

Security goal! - confidentiality protects information from unauthorized access.

Real-world Example! - An attackey intercepts data on an unsecured public wifi network.

HTTP/ TLS encryption helps prevent the attacker from reading the information.

2. Data Tampering → Unauthorized modification → Incorrect Data → Integrity.

Attacker!

Threat! An attacker changes data without permission.

Threat! The original information becomes inaccurate or unreliable.

RISK! - Wrong decisions, financial loss or system errors.

security goal! - Integrity ensures that data is accurate and has not been improperly changed.

Denial-of-service (DOS) → Service disruption → Unavailability RISK → Availability.

Attack:- An Attacker sends a huge no'of requests to a server.

Thread:- The server becomes overloaded.

Additional Links:-

Security Attack	Threat	RISK	Security Goal	Real-world Example.
phishing	credential theft	Account takeover	confidentiality	Fake banking email steals login details.
malware / Ransomware	Data destruction/ encryption	Data loss and business interruption	Integrity + availability	Ransomware encrypts hospital files.
man-in-the-middle	communication/ modification	Data exposure or alteration	confidentiality + integrity	Attacker intercepts an online transaction.
password Attack	unauthorized access	Account compromise	confidentiality	Brute-force attack guess a weak password.

Simple relationship to remainder,-

Attack → create Thread → cause RISK → security Goal reduces/ control the risk.

Attack

↓

Threat

↓

Risk

↓

Security Goal

↓

protection.

CIA Triad:-

* confidentiality:- prevent unauthorized disclosure

* integrity → prevent unauthorized modification.

* Availability:- keep service available.

oncept map : classical Encryption schemas:-

Classical Encryption Schemas	
Substitution :- Replaces plaintext symbols with other symbols	Transporation :- Rearrangs plaintext symbol without changing them.
xamples:- x caesar cipher x monoalphabetic cipher x vigenere cipher Hill cipher	Example: * Rail force cipher x Columnar Transporation.
ata structures:- Symbols change: ositions generaly remains the same. urity :- weak for classical methods	Data Structures :- Symbols remains the same positions change. security: weak when used alone.

imilarities:- Both uses keys, convert plaintext into ciphertext, and im to provide confidenticuity.

Differences and Similarities:-

ature	Substitution Technique	Transporation Technique.
sic eration	Replaces Symbols	Rearranges Symbols
pacter ntity	changes	Remains the same.
racter sitions	Generaly preserved	changed
ter uency	often preserved in simple substitutions : patterns may change	presered exactly
mples	Caesar, Vigenere, Hill	Rail fence, Columar
rity	classical methods are weak against suitable attacks	classical methods are weak against suitable Attacks.

3. How each affects Data structure:-

Substitution example:-

Caesar cipher : shift = 3

HELLO

↓

KHOOR

The letters change, but their order stays the same.

Transporation example:-

Illustrative Example rearrangement

HELLO

↓

LHOEL

The same letters remain, but their positions change.